ΣMERTXE

MEGHA H R

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CAREER OBJECTIVE

To build a career as an Embedded Systems Engineer by applying my knowledge of Embedded C, Data Structures, Microcontrollers, and electronics fundamentals. Seeking an opportunity to work on real-time embedded systems and firmware development, while continuously learning new technologies and contributing effectively to the organization.

WORK EXPERIENCE

Embedded Systems Trainee – Emertxe Information Technologies, Bengaluru 2026 – Present

- Currently undergoing an Advanced Embedded Systems Course focused on Embedded C, Data Structures, C++ and embedded software development.
- Gaining hands-on experience in Embedded C programming, Data Structures, Linux, C++, and firmware development concepts.
- Developing and implementing programming projects as part of the technical training, including Address Book Management System, Steganography, Arbitrary Precision Calculator, and Data Structures applications.
- Strengthening problem-solving, debugging, and programming skills through practical assignments and project-based learning.

TECHNICAL SKILLS

- Programming Languages:
  - C Programming
  - Advanced C programming
  - OOP using C++
  - Data structures
- System programming:
  - Linux Fundamentals
  - Linux Shell Commands
  - File Handling
  - System Calls
  - Process Management
- C Programming Concepts:
  - Pointers
  - Dynamic Memory Allocation
  - Structures and Unions
  - Strings and Arrays
  - Bitwise Operations

    - Function Pointers
    - File I/O
    - Preprocessor and Macros
- Development tools:
    - Linux / Ubuntu
    - GCC / G++
    - Git and GitHub
    - VS Code
- Core Skills:
    - Problem Solving
    - Debugging
    - Algorithm Development
    - Command-Line Programming

COURSE WORK

- Microprocessor
- Digital Electronics
- Digital Signal Processing
- Embedded systems Fundamentals

EDUCATION

- B.E (ECE), VTU, 69.3%, 2022-2026
- Class – XII, 68.6%, 2022
- Class – X, 80.96%, 2020

CONTRIBUTIONS AND ACHEIVEMENTS

- Currently undergoing technical training in **Advanced Embedded Systems** at Emertxe Information Technologies, Bangalore.
- Developed projects using **Advanced C, Data Structures, Linux and system programming concepts** as part of technical training.
- Worked on projects including **Address Book Management System, Steganography, MP3 Tag Reader/Editor, and Arbitrary Precision Calculator (APC)**.

PROJECTS AT EMERTXE

Project Number:1

Title	**Address Book Management System**
Project brief	Developed a command-line based Address Book Management System using C programming to efficiently manage contact information. The application supports adding, searching, editing, deleting, and displaying contacts, with file handling used to store and retrieve contact data.
Technologies used	C Programming, Structures, Arrays, Pointers, Strings, File Handling, Functions, Command-Line Arguments
GitHub URL	https://github.com/hrm49376-megha/Address-Book-Management-System.git
Key challenges & Learnings	✓ Implemented contact management operations using structures, arrays, pointers, and functions. ✓ Developed input validation for names, mobile numbers, and email addresses. ✓ Implemented duplicate checking for mobile numbers and email IDs. ✓ Learned to handle contact data using file operations for persistent storage. ✓ Improved debugging, logical thinking, and problem-solving skills while developing and testing the application.

Project number:2

Title	**Steganography**
Project brief	Developed a Steganography application using C to hide secret text data inside a BMP image using **Least Significant Bit (LSB) encoding**. The application supports encoding secret data into an image and decoding the hidden message from the encoded image.
Technologies used	C Programming, File Handling, Pointers, Bitwise Operations, Functions, Structures, Command-Line Arguments
GitHub URL	https://github.com/hrm49376-megha/Steganography.git
Key challenges & Learnings	✓ Understood the BMP file structure and header information required for encoding and decoding. ✓ Implemented LSB-based bit manipulation to encode and decode secret data. ✓ Learned to perform binary file read and write operations. ✓ Handled secret data encoding and decoding while preserving the image properties. ✓ Improved understanding of pointers, bitwise operations, file handling, debugging, and problem-solving.

Project number:3

Title	**MP3 Tag Reader/Editor**
Project brief	Developed an MP3 Tag Reader/Editor using C programming to read and modify metadata information stored in MP3 files. The application allows users to view and edit tag information such as song title, artist, album, year, genre, and comments through command-line operations.
Technologies used	C Programming, File Handling, Structures, Pointers, Strings, Command-Line Arguments
GitHub URL	https://github.com/hrm49376-megha/MP3-Tag-Reader-Editor.git
Key challenges & Learnings	✓ Understood the structure of MP3 files and ID3 tag metadata. ✓ Implemented file operations to read and modify MP3 tag information. ✓ Used structures, pointers, and string handling to manage tag data. ✓ Implemented command-line arguments for different reader and editor operations. ✓ Improved understanding of binary file handling, debugging, and problem-solving through project implementation.

Project number:4

Title	**Arbitrary Precision Calculator (APC)**
Project brief	Developed an Arbitrary Precision Calculator using **Doubly Linked Lists** to perform arithmetic operations on very large numbers that cannot be handled by standard C data types. The application supports addition, subtraction, multiplication, and division through command-line arguments.
Technologies used	C Programming, Doubly Linked Lists, Pointers, Dynamic Memory Allocation, Functions, Command-Line Arguments
GitHub URL	https://github.com/hrm49376-megha/Arbitrary-Precision-Calculator-APC-.git
Key challenges & Learnings	✓ Implemented large-number storage using doubly linked lists instead of standard integer data types. ✓ Developed arithmetic operations including addition, subtraction, multiplication, and division. ✓ Handled positive and negative numbers and different input cases during calculations. ✓ Managed dynamic memory allocation and linked-list traversal for large numbers. ✓ Improved understanding of data structures, pointers, dynamic memory management, debugging, and algorithmic problem-solving.

ACADEMIC PROJECTS

Title	**Design and Development of Traction Inverter for AC Induction Motor**
Project brief	Designed and developed a **three-phase Space Vector Pulse Width Modulation (SVPWM) based DC-to-variable-frequency AC inverter** for driving an AC induction motor in electric vehicle traction applications. The system converts DC power into controlled three-phase AC output with variable voltage and frequency for motor operation.
Technologies used	Arduino, SVPWM, Power Electronics, AC Induction Motor, Gate Driver Circuit, Voltage and Current Sensors
Key challenges & Learnings	✓ Studied the working principles of three-phase inverters and AC induction motors. ✓ Implemented SVPWM concepts for generating controlled switching pulses using Arduino. ✓ Gained knowledge of gate driver circuits and power semiconductor switching. ✓ Worked with voltage and current sensing concepts for monitoring the inverter system. ✓ Improved understanding of motor control, power electronics, circuit analysis, and troubleshooting.